

The term *artificial intelligence* has an aura about it that can make people think that it's mysterious and highly advanced. Some believe that AI enables computers to think, feel, and experience conscious thought in the same way that humans do; whether computers will ever be able to do so is an open, difficult question that is far beyond the scope of this chapter. The AI that we'll build is much simpler and will be capable of playing a game well, but not of writing sincerely felt love poems or feeling despondency or desire (as far as I can tell!).

Our AI will be able to play *dots and boxes*, a simple but nontrivial game played worldwide. We'll start by drawing the game board. Then we'll build functions to keep score as games are in progress. Next, we'll generate game trees that represent all possible combinations of moves that can be played in a given game. Finally, we'll introduce the minimax algorithm, an elegant way to implement AI in just a few lines.

La Pipopipette

Dots and boxes was invented by the French mathematician Édouard Lucas, who named it *la pipopipette*. It starts with a *lattice*, or grid of points, like the one shown in [Figure 10-1](#).